STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: WLD-01 | TAG: TIG-275-01

WLD-01: Lincoln Precision TIG 275 Welder #1

Facility Zone: Zone 3: Welding & Thermal Processing (WLD) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses Under Shade 10-13 Auto-Darkening Helmet
2. Dry Hole-Free Leather TIG Gloves
3. Flame-Resistant (FR) Cotton/Leather Apparel
4. Active Fume Extraction Arm Positioned Within 12"

2. PRIMARY OPERATIONAL HAZARDS

1. Intense UV/IR arc radiation (arc eye & skin burns)
2. High-frequency (HF) start electrical shock hazard
3. 2,500 PSI compressed gas cylinder hazard & Argon asphyxiation in confined spaces

3. BEFORE OPERATION CHECKLIST

1. Inspect torch collet, gas lens, ceramic cup, and ground tungsten electrode.
2. Verify work lead (ground clamp) is securely clamped to clean bare metal.
3. Turn on Argon cylinder valve slowly and set flow rate (15-25 CFH).

4. DURING OPERATION SAFETY RULES

1. Keep welding curtains closed to protect bystanders from arc flash.
2. Observe machine duty cycle ratings and keep torch cable away from hot weldment.

5. AFTER USE & SHOWROOM CLEANUP

1. Allow post-flow Argon to cool tungsten before lifting torch.
2. Close Argon bottle valve completely (all yellow), purge line via foot pedal, and power machine OFF.
3. Coil torch and ground leads neatly on rack.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Release foot pedal immediately, switch main power OFF, close Argon cylinder valve, and alert Lab Supervisor.